

Introduction to Image Processing

Practical Work No. 1

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Link to the programming code: <https://github.com/ibrahimegebilgic/practical1-imageprocessing>

Variant: 4 (Multiply, Color Dodge, Darken, Color Burn)

```
In [1]: import os
import numpy as np
from PIL import Image, ImageDraw, ImageFont
import matplotlib.pyplot as plt
import urllib.request
```

```
In [2]: def load_image(path):
        return Image.open(path).convert("RGB")

URL_A = "https://images.unsplash.com/photo-1771272338329-1ab5145559a5?auto=format&f
URL_B = "https://plus.unsplash.com/premium_photo-1672201106204-58e9af7a2888?auto=fo

def download_image(url: str, out_path: str) -> bool:
    try:
        urllib.request.urlretrieve(url, out_path)
        return True
    except Exception as e:
        print(f"Download failed for {out_path}: {e}")
        return False

def get_images(size=512):
    outA = "image_a.jpg"
    outB = "image_b.jpg"

    download_image(URL_A, outA)
    download_image(URL_B, outB)

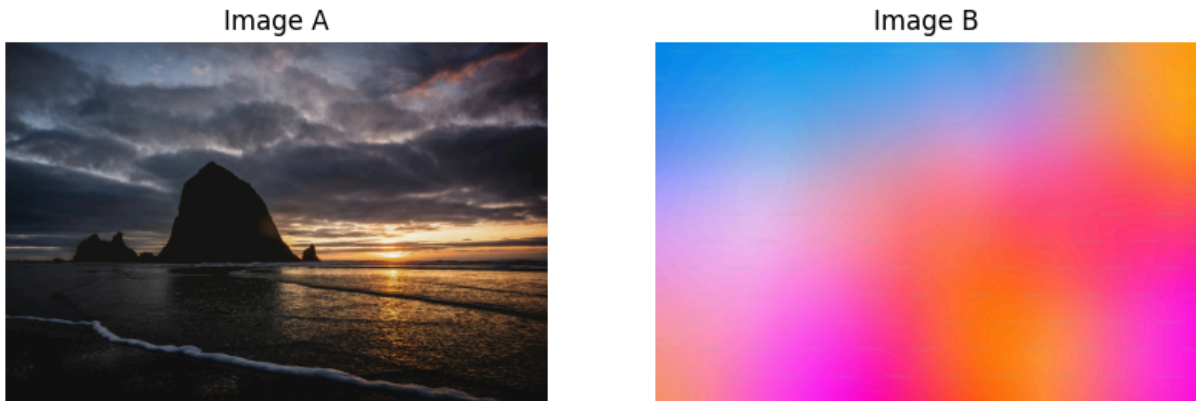
    imgA = load_image(outA)
    imgB = load_image(outB)

    return imgA, imgB

imgA, imgB = get_images(512)
imgA.size, imgB.size
```

Out[2]: ((1200, 800), (1200, 800))

```
In [3]: fig, ax = plt.subplots(1, 2, figsize=(10, 5))
ax[0].imshow(imgA)
ax[0].set_title("Image A")
ax[0].axis("off")
ax[1].imshow(imgB)
ax[1].set_title("Image B")
ax[1].axis("off")
plt.show()
```



```
In [4]: def to_float01(img):
        return np.asarray(img).astype(np.float32) / 255.0

def to_uint8(arr01):
    return (np.clip(arr01, 0, 1) * 255.0 + 0.5).astype(np.uint8)

def blend_multiply(a, b):
    return a * b

def blend_darken(a, b):
    return np.minimum(a, b)

def blend_color_dodge(a, b, eps=1e-6):
    denom = 1.0 - b
    res = a / np.maximum(denom, eps)
    res = np.where(b >= 1.0 - eps, 1.0, res)
    return np.clip(res, 0, 1)

def blend_color_burn(a, b, eps=1e-6):
    res = 1.0 - (1.0 - a) / np.maximum(b, eps)
    res = np.where(b <= eps, 0.0, res)
    return np.clip(res, 0, 1)
```

```
In [5]: A = to_float01(imgA)
B = to_float01(imgB)

results = {
    "Multiply": blend_multiply(A, B),
    "Color Dodge": blend_color_dodge(A, B),
    "Darken": blend_darken(A, B),
    "Color Burn": blend_color_burn(A, B),
}
```

```

}

fig, ax = plt.subplots(2, 2, figsize=(12, 10))
ax = ax.ravel()
for i, (name, arr) in enumerate(results.items()):
    ax[i].imshow(to_uint8(arr))
    ax[i].set_title(name)
    ax[i].axis("off")
plt.tight_layout()
plt.show()

```

Multiply



Color Dodge



Darken



Color Burn



```

In [6]: out_dir = "pw1_outputs"
os.makedirs(out_dir, exist_ok=True)
Image.fromarray(to_uint8(A)).save(os.path.join(out_dir, "base_A.png"))
Image.fromarray(to_uint8(B)).save(os.path.join(out_dir, "base_B.png"))
for name, arr in results.items():
    fname = "blend_" + name.lower().replace(" ", "_") + ".png"
    Image.fromarray(to_uint8(arr)).save(os.path.join(out_dir, fname))
sorted(os.listdir(out_dir))

```

```

Out[6]: ['base_A.png',
'base_B.png',
'blend_color_burn.png',
'blend_color_dodge.png',
'blend_darken.png',
'blend_multiply.png']

```

```

In [8]: !pip install nbconvert

```

Requirement already satisfied: nbconvert in /usr/local/lib/python3.12/dist-packages (7.17.0)

Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (4.13.5)

Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert) (6.3.0)

Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.7.1)

Requirement already satisfied: Jinja2>=3.0 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.1.6)

Requirement already satisfied: jupyter-core>=4.7 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.9.1)

Requirement already satisfied: jupyterlab-pygments in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.3.0)

Requirement already satisfied: markupsafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.0.3)

Requirement already satisfied: mistune<4,>=2.0.3 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.2.0)

Requirement already satisfied: nbclient>=0.5.0 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.10.4)

Requirement already satisfied: nbformat>=5.7 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.10.4)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from nbconvert) (26.0)

Requirement already satisfied: pandocfilters>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (1.5.1)

Requirement already satisfied: pygments>=2.4.1 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (2.19.2)

Requirement already satisfied: traitlets>=5.1 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.7.1)

Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert) (0.5.1)

Requirement already satisfied: tinycss2<1.5,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert) (1.4.0)

Requirement already satisfied: platformdirs>=2.5 in /usr/local/lib/python3.12/dist-packages (from jupyter-core>=4.7->nbconvert) (4.9.2)

Requirement already satisfied: jupyter-client>=6.1.12 in /usr/local/lib/python3.12/dist-packages (from nbclient>=0.5.0->nbconvert) (7.4.9)

Requirement already satisfied: fastjsonschema>=2.15 in /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (2.21.2)

Requirement already satisfied: jsonschema>=2.6 in /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (4.26.0)

Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert) (2.8.3)

Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert) (4.15.0)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (25.4.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.37.0)

Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.30.0)

Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-packages

```
s (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (0.4)
Requirement already satisfied: nest-asyncio>=1.5.4 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.6.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (2.9.0.post0)
Requirement already satisfied: pyzmq>=23.0 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (26.2.1)
Requirement already satisfied: tornado>=6.2 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (6.5.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.17.0)
```

```
In [10]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [11]: !jupyter nbconvert "/content/drive/MyDrive/Colab Notebooks/Practical_1.ipynb" \
--to html \
--embed-images
```

```
[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab Notebooks/Practical_1.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 5 image(s).
[NbConvertApp] Writing 3774496 bytes to /content/drive/MyDrive/Colab Notebooks/Practical_1.html
```

```
In [12]: from google.colab import files
files.download("/content/drive/MyDrive/Colab Notebooks/Practical_1.html")
```

```
In [13]: import os
for root, dirs, files in os.walk('/content/drive/MyDrive/'):
    for name in files:
        if 'Praktiskais darbs 1.ipynb' in name:
            print(os.path.join(root, name))
```

/content/drive/MyDrive/Colab Notebooks/Praktiskais darbs 1.ipynb

Conclusions

I developed four distinct blending modes: Multiply, Color Dodge, Darken, and Color Burn. Instead of relying on pre-existing functions, I utilized mathematical equations with the help of the NumPy and OpenCV libraries. I tested a variety of portrait and high-contrast images to verify my calculations. I also created crossfade and wipe transition effects for showcasing the smooth movement of images. I've concluded that each blending mode alters the brightness and contrast of the resulting image differently.